In [5]:
```
#import lib

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

Q1: Loading and Understanding the Dataset

In [10]:
```
#Q1.1: Load dataset

df = sns.load_dataset("tips")
```

In [11]:
```
#Q1.2: Display first rows

df.head()
```

Out[11]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [12]:
```
#Q1.3: Dataset shape and columns

print(df.shape)
print(df.columns)
```

```
(244, 7)
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

Q2: Basic Filtering – Who Tips More?

In [21]:
```
#Q2.1: Weekend
weekend_customers = df[ (df["day"] == "Sat") | (df["day"] == "Sun") ]
avg_tip_weekend = weekend_customers["tip"].mean()

# weekday
weekday_customers = df[ (df["day"] != "Sat") & (df["day"] != "Sun") ]
avg_tip_weekday = weekday_customers["tip"].mean()

avg_tip_overall = df["tip"].mean()

print(avg_tip_weekend)
```

```
print(avg_tip_weekday)
print(avg_tip_overall)

if avg_tip_weekend > avg_tip_weekday:
    print("Conclusion: Weekend customers leave higher average tips.")
else:
    print("Conclusion: Weekday customers leave higher average tips.")
```

```
3.115276073619632
2.7628395061728392
2.99827868852459
Conclusion: Weekend customers leave higher average tips.
```

Q3: Sorting and Identifying High-Value Customers

In [30]:
```
# Q3.1: Sort customers by total_bill (descending).
df.sort_values("total_bill", ascending = False)

# Q3.2:Display the top 10 highest bills.
df.sort_values("total_bill", ascending = False).head(5)

# Analys:
# Most of the top 10 highest bills are associated with larger group sizes, typicall
# This suggests that higher total bills are mainly driven by larger groups rather t
```

Out[30]:

	total_bill	tip	sex	smoker	day	time	size
170	50.81	10.00	Male	Yes	Sat	Dinner	3
212	48.33	9.00	Male	No	Sat	Dinner	4
59	48.27	6.73	Male	No	Sat	Dinner	4
156	48.17	5.00	Male	No	Sun	Dinner	6
182	45.35	3.50	Male	Yes	Sun	Dinner	3

Q4: Data Quality Check & Cleaning

In [31]:
```
#Q4.1: missing values
df.isna().sum()

# Missing values are dangerous in machine learning because many algorithms cannot h
# They can lead to incorrect model training, biased predictions, and loss of valuab
```

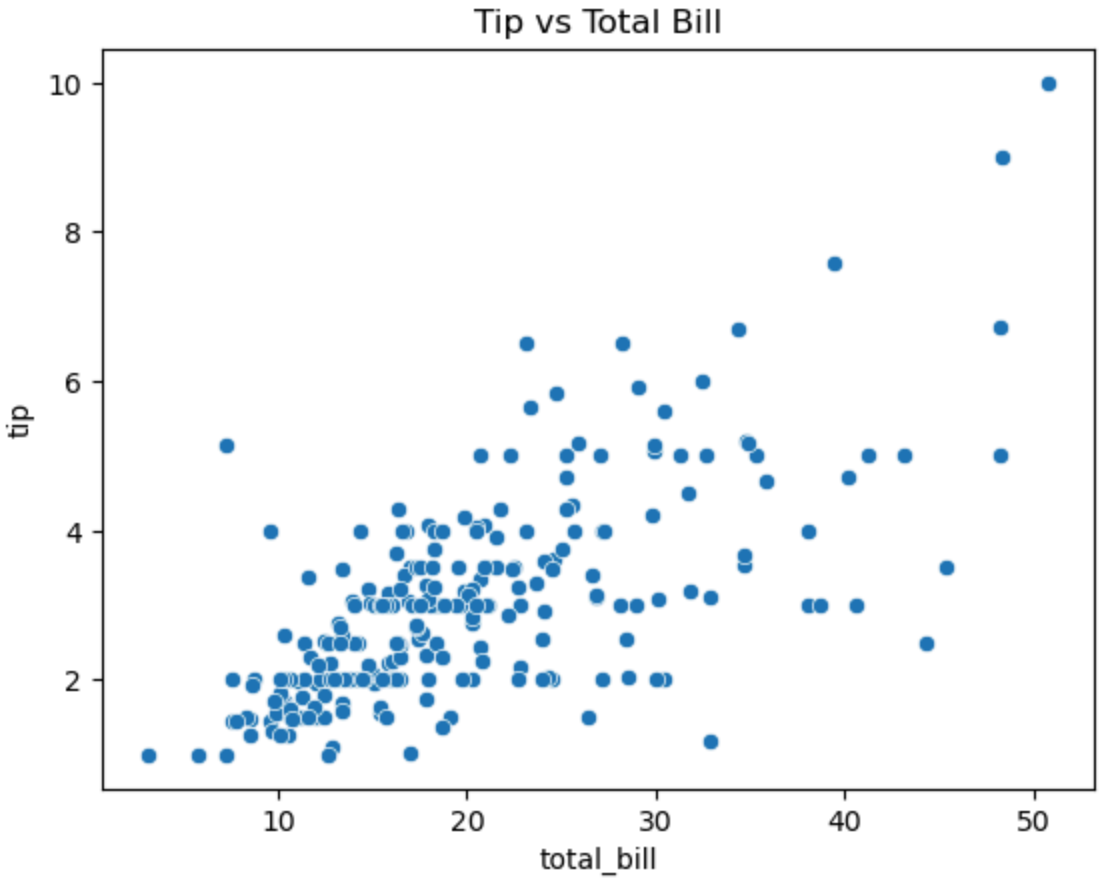
Out[31]:
```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64
```

Q5: Visualization – What Influences Tips?

In [37]:
```
#Q5.1: Tip vs Total Bill (scatter plot)

plt.title("Tip vs Total Bill")
sns.scatterplot(x="total_bill", y="tip", data=df)
plt.show()

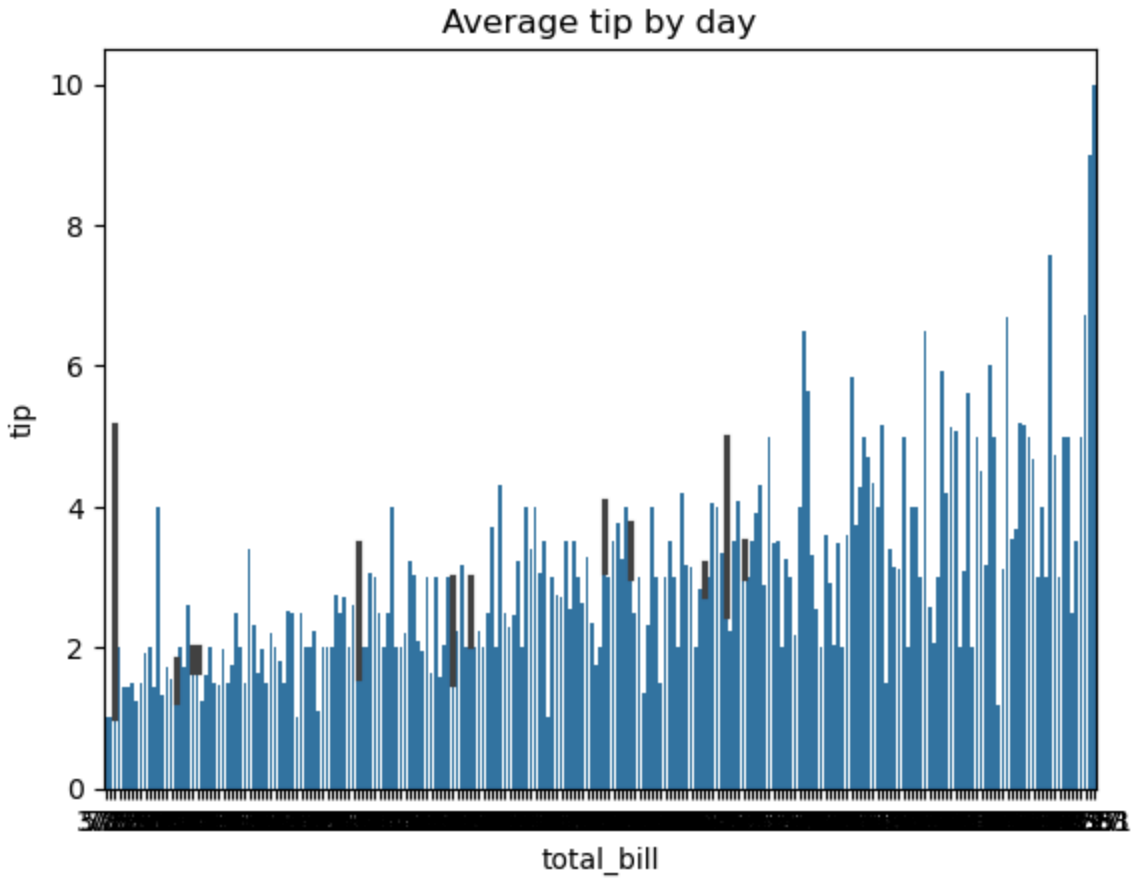
# The scatter plot shows a positive relationship between total bill and tip amount.
# As the total bill increases, customers generally tend to leave higher tips.
```



In [41]:
```
# Q5.1: Average tip by day (bar plot)

plt.title("Average tip by day")
sns.barplot(x="total_bill", y="tip", data=df, estimator = "mean")
plt.show()

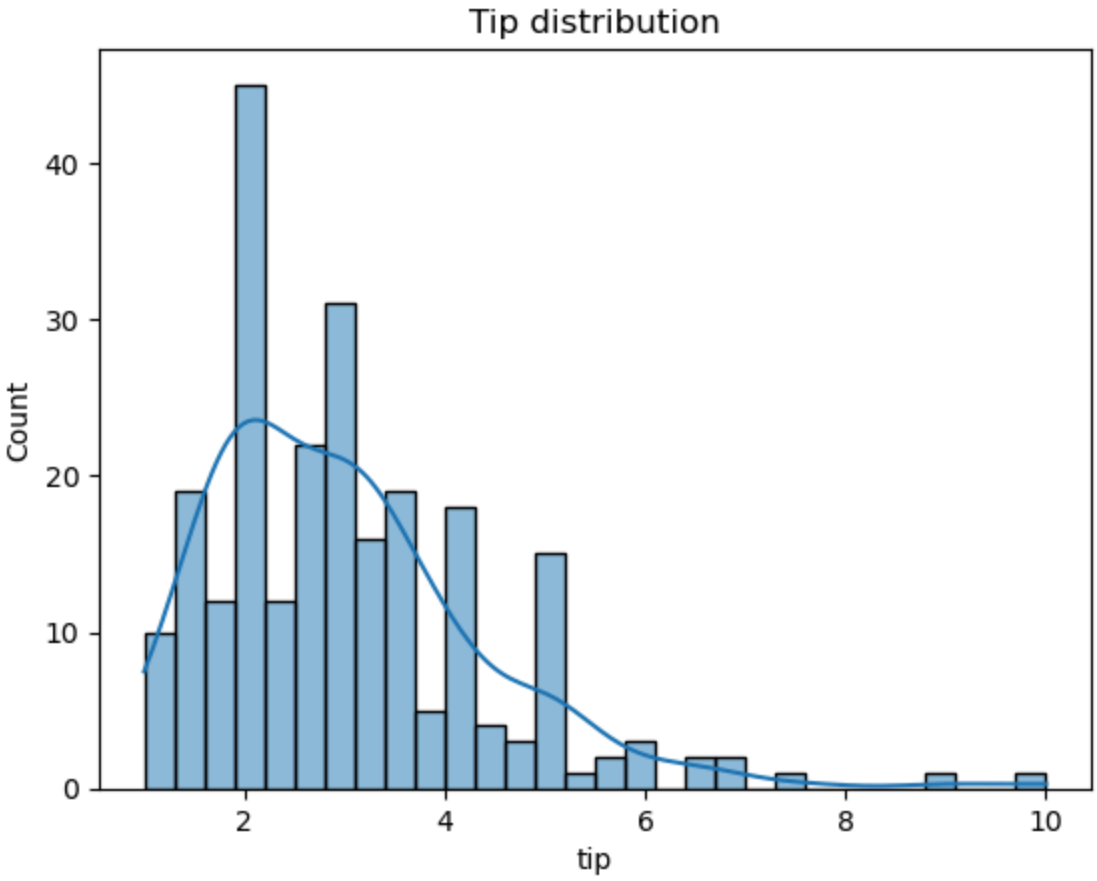
# The bar plot shows that average tips vary by day.
# Weekend days, especially Saturday and Sunday, tend to have slightly higher averag
```



In [47]:
```
# Q5.1: Tip distribution (histogram)

plt.title("Tip distribution")
sns.histplot(x = "tip", data = df, bins = 30, kde = True)
plt.show()

# The histogram shows that most tips are concentrated at lower values.
# Higher tips are less frequent, indicating a right-skewed distribution.
```



In [48]:
```
# Tip amount is influenced by the total bill size and the day of visit.
# Higher bills and weekend visits are generally associated with higher tips, while
```

Q6: Final Insight & Conclusion

Customers:

Customers generally leave higher tips when the total bill is higher. Weekend customers, especially on Saturday and Sunday, tend to tip more than weekday customers. Tips are also higher during dinner time, which often involves larger groups and higher spending.

Restaurant Managers:

These findings show that weekends and dinner hours are important for restaurant revenue. Managers can schedule more staff during these busy periods to provide better service. Focusing on weekend customers and larger groups can help increase both sales and tips.

In [ ]:
```

```